QUBO Formulation: Traveling Salesperson Problem

Traveling Salesperson Problem

Problem Statement. Given a complete directed graph $G = (V, E)$ with $|V| = N$ cities and a non-negative distance matrix $D = (d_{ij})_{i,j \in V}$, find a Hamiltonian cycle that visits every city exactly once and returns to the starting city, such that the total travel distance is minimized.

Decision Variables. For every ordered pair of distinct cities (i, j) with $i, j \in V$ and $i \neq j$, we introduce a binary decision variable $x_{ij} \in \{0, 1\}$. The variable x_{ij} equals 1 if the directed edge (i, j) is included in the tour, and 0 otherwise.

QUBO Derivation. We construct a Quadratic Unconstrained Binary Optimization (QUBO) Hamiltonian $H(\mathbf{x})$ by combining the objective function with quadratic penalty terms that enforce the problem constraints. Let $\lambda > 0$ be a penalty weight parameter.

Step 1 – Objective Function. The original minimization objective is the total travel distance:

$$O(\mathbf{x}) = \sum_{i \in V} \sum_{j \in V, j \neq i} d_{ij} x_{ij}. \tag{1}$$

Step 2 – Constraint Formulation. The problem requires three sets of constraints:

$$\text{(Out-degree)} \quad \sum_{j \in V, j \neq i} x_{ij} = 1, \quad \forall i \in V, \tag{2}$$

$$\text{(In-degree)} \quad \sum_{i \in V, i \neq j} x_{ij} = 1, \quad \forall j \in V, \tag{3}$$

$$\text{(Subtour elimination)} \quad \sum_{i \in S} \sum_{j \in S, j \neq i} x_{ij} \leq |S| - 1, \quad \forall S \subset V,\ 2 \leq |S| \leq N - 1. \tag{4}$$

Step 3 – Penalty Term Expansion. Each equality constraint is converted to a quadratic penalty. For the out-degree constraint associated with city i, we write:

$$P_i^{\text{out}}(\mathbf{x}) = \lambda \left(\sum_{j \neq i} x_{ij} - 1 \right)^2. \tag{5}$$

Expanding the square and applying the idempotent property $x_{ij}^2 = x_{ij}$ for binary variables yields:

$$\left(\sum_{j\neq i} x_{ij} - 1\right)^2 = \sum_{j\neq i} x_{ij}^2 + \sum_{j\neq k, j,k\neq i} x_{ij}x_{ik} + 1 - 2\sum_{j\neq i} x_{ij} \tag{6}$$

$$= 1 - \sum_{j\neq i} x_{ij} + \sum_{j\neq k, j,k\neq i} x_{ij}x_{ik}. \tag{7}$$

Summing over all $i \in V$, the total out-degree penalty contribution is:

$$P^{\text{out}}(\mathbf{x}) = \lambda \sum_{i\in V} \left(1 - \sum_{j\neq i} x_{ij} + \sum_{j\neq k, j,k\neq i} x_{ij}x_{ik}\right). \tag{8}$$

Analogously, the in-degree penalty for city j expands to:

$$P^{\text{in}}(\mathbf{x}) = \lambda \sum_{j\in V} \left(1 - \sum_{i\neq j} x_{ij} + \sum_{i\neq k, i,k\neq j} x_{ij}x_{kj}\right). \tag{9}$$

The subtour elimination constraints are enforced via a similar quadratic penalty over all valid subsets S:

$$P^{\text{sub}}(\mathbf{x}) = \lambda \sum_{\substack{S\subset V \\ 2\leq |S|\leq N-1}} \left(\sum_{i\in S}\sum_{j\in S, j\neq i} x_{ij} - (|S|-1)\right)^2. \tag{10}$$

Step 4 – Combined Hamiltonian. The full QUBO Hamiltonian is the sum of the objective and all penalty terms:

$$H(\mathbf{x}) = O(\mathbf{x}) + P^{\text{out}}(\mathbf{x}) + P^{\text{in}}(\mathbf{x}) + P^{\text{sub}}(\mathbf{x}). \tag{11}$$

Step 5 – Q Matrix Entries and Offset. Collecting linear, quadratic, and constant terms from the expanded form above, the QUBO parameters are given in closed form as:

$$Q_{ij,ij} = d_{ij} - \lambda, \quad \forall i \neq j, \tag{12}$$

$$Q_{ij,ik} = 2\lambda, \quad \forall i \in V,\ j \neq k, \tag{13}$$

$$Q_{ji,jk} = 2\lambda, \quad \forall j \in V,\ i \neq k, \tag{14}$$

$$Q_{ij,kl} = 2\lambda, \quad \text{for pairs appearing in subtour subsets } S, \tag{15}$$

$$c = 2N\lambda + \lambda \sum_{\substack{S\subset V \\ 2\leq |S|\leq N-1}} (|S|-1)^2. \tag{16}$$

The Hamiltonian takes the standard QUBO form $H(\mathbf{x}) = \sum_i Q_{ii}x_i + \sum_{i<j} Q_{ij}x_ix_j + c$.

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any constraint violation is strictly more costly than the maximum possible reduction in the objective function. Let $D_{\max} = \max_{i,j \in V} d_{ij}$. If a single degree constraint is violated, the penalty increases by at least λ. The maximum possible decrease in the objective function from any single variable flip is bounded by $D_{\max}$. Therefore, choosing $\lambda > D_{\max}$ guarantees that the energy cost of violating a constraint always outweighs any potential gain in the objective, forcing the optimizer toward feasible configurations.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\mathbf{x} \in \{0,1\}^{N(N-1)}$ recovers the optimal TSP tour because the Hamiltonian is constructed such that $H(\mathbf{x}) = O(\mathbf{x})$ for all feasible $\mathbf{x}$ (where all constraints are satisfied and penalties vanish), and $H(\mathbf{x}) > O(\mathbf{x})$ for all infeasible $\mathbf{x}$. Since $\lambda > D_{\max}$, any infeasible configuration has an energy strictly greater than the best feasible configuration. Consequently, the global minimum of $H(\mathbf{x})$ must occur at a feasible point, and among feasible points, $H(\mathbf{x})$ reduces exactly to the objective function $O(\mathbf{x})$. Thus, the global minimizer corresponds precisely to the minimum-distance Hamiltonian cycle.

Illustrative Example. Consider a concrete instance with $N = 3$ cities, $V = \{0, 1, 2\}$. The directed edge variables are $x_{01}, x_{02}, x_{10}, x_{12}, x_{20}, x_{21}$. The objective is:

$$O(\mathbf{x}) = d_{01}x_{01} + d_{02}x_{02} + d_{10}x_{10} + d_{12}x_{12} + d_{20}x_{20} + d_{21}x_{21}. \tag{17}$$

The out-degree and in-degree penalties expand to:

$$P^{\text{out}} = \lambda \left[(1 - x_{01} - x_{02} + x_{01}x_{02}) + (1 - x_{10} - x_{12} + x_{10}x_{12}) + (1 - x_{20} - x_{21} + x_{20}x_{21}) \right], \tag{18}$$

$$P^{\text{in}} = \lambda \left[(1 - x_{10} - x_{20} + x_{10}x_{20}) + (1 - x_{01} - x_{21} + x_{01}x_{21}) + (1 - x_{02} - x_{12} + x_{02}x_{12}) \right]. \tag{19}$$

For $N = 3$, the subtour elimination constraint applies only to $S = \{0, 1, 2\}$, yielding a single penalty term:

$$P^{\text{sub}} = \lambda \left(x_{01} + x_{02} + x_{10} + x_{12} + x_{20} + x_{21} - 2 \right)^2. \tag{20}$$

Combining all terms, the concrete Hamiltonian is:

$$H(\mathbf{x}) = \sum_{i \neq j} (d_{ij} - \lambda) x_{ij} + 2\lambda \sum_{\text{quadratic pairs}} x_{ij}x_{kl} + 6\lambda. \tag{21}$$

For a symmetric distance matrix with $d_{ij} = 1$ for all $i \neq j$ and $\lambda = 2$, the optimal tour is $0 \to 1 \to 2 \to 0$ (or its reverse), corresponding to $x_{01} = x_{12} = x_{20} = 1$ and all other $x_{ij} = 0$. The objective value is 3, the penalty terms evaluate to 0, and the total energy is $H^* = 3$. Any infeasible assignment (e.g., $x_{01} = x_{10} = 1$, others 0) incurs a penalty of at least $2\lambda = 4$, resulting in $H \geq 4 + 2 = 6$, confirming that the global minimum correctly identifies the valid tour.